IoT-DriftGen: Horizon-Based Early Warning for Predictive Maintenance with Drift-Aware Temporal Evaluation

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***Abstract*—Predictive maintenance (PdM) systems are commonly evaluated as reactive classifiers that detect failure at the current timestep, yet industrial practice requires actionable early warning over a future horizon. This mismatch is compounded when randomly stratified evaluation mixes temporally adjacent windows across partitions, potentially inflating performance estimates.**

We present *IoT-DriftGen*, a horizon-based PdM framework whose central contribution is methodological: we show that the operationally meaningful metric—lead time per failure event—is robust to evaluation protocol, even when raw classification AUC is not. To make this argument concrete we introduce FTEPH (Failure-Type Early Prediction Horizon), an event-level metric quantifying actionable lead time per failure mode, and conduct a stratified-vs-chronological sensitivity analysis across two independent benchmarks. The detection backbone combines physics-informed feature engineering with sliding-window temporal statistics and a horizon-aware XGBoost classifier. A compact TFT-inspired quantile forecaster augments the pipeline with uncertainty-aware sensor trajectories for decision-support simulation.

Under chronological evaluation FTEPH mean lead-time changes by less than one cycle on both benchmarks (AI4I 2020: 33.2 to 33.0 cycles; CMAPSS FD001: 46.2 to 46.5 cycles), even though raw classification AUC drops by 22 points on AI4I and 0.7 points on CMAPSS. Multi-seed experiments (5 seeds) confirm stable detection with ROC-AUC $= 0.9868 \pm 0.0052$ (95% CI), and component ablation identifies temporal sliding-window statistics as the dominant contributor (ΔAUC $= -0.117$ if removed).

***Index Terms*—predictive maintenance, industrial IoT, horizon-based detection, drift-aware evaluation, temporal evaluation, XGBoost, Temporal Fusion Transformer, CMAPSS, FTEPH**

I. Introduction

Unplanned equipment failures cost the manufacturing sector an estimated $50 billion per year in downtime and secondary damage [1]. Despite a decade of IIoT-enabled research, a practical gap persists between academic detection metrics and operational utility: a system that classifies failure *at the moment it happens* provides almost no actionable lead time.

Two structural problems underlie this gap. First, most methods define the prediction target at a single timestep—predicting whether the machine is failing *now*—which is too late for maintenance planning. Second, reconstruction-based detectors (e.g., β-VAEs) fail on datasets like AI4I 2020 where failures are point-like with no preceding gradual drift in raw sensor values. Third, and less discussed, randomly stratified evaluation may overestimate deployment performance by mixing temporally adjacent windows across partitions, leaking future operating regimes into training data.

This paper presents *IoT-DriftGen* with the following contributions: **(1)** *Horizon-based binary detection*—a K-step-ahead formulation that converts failure timestamps into actionable early-warning targets without requiring precise degradation labels; **(2)** *FTEPH*, an event-level metric that quantifies and compares actionable lead time per failure mode under any evaluation protocol; **(3)** a *stratified-vs-chronological sensitivity analysis* on two independent benchmarks showing that FTEPH is split-invariant even when classification AUC is not—the central methodological finding; **(4)** physics-informed feature engineering combined with sliding-window temporal statistics, ablated to quantify their relative contribution; and **(5)** a compact TFT-inspired quantile forecaster used as a decision-support simulation layer, validated against persistence and moving-average baselines.

II. Related Work

Classical supervised methods—Random Forests, gradient boosting, and SVMs—remain competitive for tabular industrial data because engineered features capture failure signatures that raw sensor values do not directly expose [2]. XGBoost [3] is particularly effective given its robustness to class imbalance and heterogeneous feature scales. LSTM [4] and Transformer architectures extend this to sequence modeling, but on small-to-medium tabular datasets engineered temporal features still dominate [5].

Unsupervised VAE-based detectors [6] require no fault examples and work well when failures involve gradual signal drift, but fail on point-like failure regimes such as AI4I 2020. Remaining Useful Life (RUL) regression [7] provides an alternative but requires precise degradation labels rarely available in industrial environments. Horizon-based binary detection—predicting whether failure occurs within a future window—is a middle ground that needs only failure timestamps and naturally

produces early-warning outputs, yet remains underexplored in the PdM literature.

For probabilistic trajectory generation, the Temporal Fusion Transformer (TFT) [8] introduced multi-horizon forecasting with calibrated uncertainty via quantile regression. We adopt a compact, adaptation of the TFT architecture as a decision-support forecasting component rather than proposing it as a novel architecture; our contribution is its integration and validation within the early-warning pipeline, not the model design itself.

III. Methodology

Datasets. Our primary benchmark is AI4I 2020 [9]: 10 000 simulated industrial records, 5 raw sensors, failure modes TWF/HDF/PWF/OSF/RNF, $\sim$3.4% failure rate. For generalization we use NASA CMAPSS FD001 [10]: 20 631 cycles, 100 turbofan units, 16 sensors, single failure mode; binary label derived from RUL as $y_t = \mathbb{1}[\mathrm{RUL}(t) \le H]$ with H=15.

Although AI4I exhibits point-like failures and CMAPSS exhibits smooth degradation trajectories, both are compatible with horizon-based detection: the K-step-ahead label (4) is defined purely from failure timestamps and is agnostic to the underlying degradation shape. The temporal sliding-window statistics then encode whatever precursor pattern the regime provides—abrupt signature changes on AI4I, gradual drift slopes on CMAPSS—without requiring the feature engineering to be redesigned per dataset.

Phase A—Physics-Informed Features. On AI4I we derive

$$\text{Power} = \text{Torque} \times \text{RotSpeed}, \tag{1}$$

$$\text{TempDiff} = \text{ProcessTemp} - \text{AirTemp}, \tag{2}$$

$$\text{ToolStrain} = \text{ToolWear} \times \text{Torque}, \tag{3}$$

plus two ratio terms and a heat-rate indicator, mapping directly to the relevant failure modes (HDF←TempDiff, PWF←Power, OSF/TWF←ToolStrain). Total: 14 channels (5 raw + 9 engineered). On CMAPSS the sensor semantics are anonymized; we retain the 16 measured channels without cross-term engineering, isolating the temporal block as the load-bearing component.

Phase B—Sliding-Window Statistics. For each channel we compute 12 statistics over W=20 cycles: mean, std, min, max, range, OLS slope, first-order difference δ, last value, last−mean, mean_abs_diff, std_diff, and max_jump. This yields 168 features on AI4I or 192 on CMAPSS; the raw (W, F) tensor is also retained for the TFT.

Phase C—Horizon-Based Binary Detection. The central contribution replaces the standard label $y_t = \mathbb{1}[t = t_f]$ with the horizon label

$$y_t^{(K)} = \mathbb{1}\big[\exists\, t_f \in [t{+}1,\ t{+}K]\big], \quad K = 10. \tag{4}$$

We train an XGBoost classifier [3] with `scale_pos_weight` = neg/pos and tune nine hyperparameters via 60 Optuna [11] TPE trials optimizing AUCPR under 5-fold stratified cross-validation. The operating threshold is selected by an F1-maximizing sweep on the held-out validation set before final test evaluation.

Phase D—FTEPH. For each failure event e occurring at $t_f^{(e)}$, the detection horizon is $h(e) = t_f^{(e)} - \min\{t : \hat{y}_t^{(K)}{=}1,\ t \in [t_f^{(e)}{-}L,\ t_f^{(e)})\}$ with causal lookback $L{=}5K$. FTEPH at threshold $h_{\min}$ for type c is the fraction of events in c with $h(e) \ge h_{\min}$.

Phase E—Diagnosis. A second XGBoost is trained on failure-only AI4I samples (RNF excluded as definitionally stochastic) to predict the failure mode in $\{\mathrm{HDF}, \mathrm{PWF}, \mathrm{OSF}, \mathrm{TWF}\}$.

Phase F—TFT Quantile Forecaster. We integrate a compact adaptation of the TFT architecture [8] comprising 509 636 parameters: sinusoidal positional encoding, four-head self-attention (d=96), Gated Linear Unit encoder, and a learned failure-type embedding conditioning the decoder. The decoder outputs $q_{10}/q_{50}/q_{90}$ for each of the next K=10 steps and each feature. Training uses pinball loss $\mathcal{L} = \sum_q \sum_k \sum_f \rho_q(x_{t+k,f} - \hat{x}^{(q)}_{t+k,f})$ with feature-scaled jitter and manifold mixup (early stopping: patience 20, max 200 epochs).

Phase G—Decision-Support Simulation. Starting from a true-positive seed window, at each step the simulation (i) scores the state with XGBoost, (ii) generates the next observation with the TFT q_{50}, (iii) shifts the window forward. Trajectory consistency is

$$\text{Consistency}(c) = \frac{\overline{\text{risk}}_c - \text{baseline}_{\text{healthy}}}{1 - \text{baseline}_{\text{healthy}}}, \tag{5}$$

where the healthy baseline is the mean detector risk on confirmed negative test windows. Consistency > 0 indicates that the failure-type embedding is associated with elevated simulated risk relative to the healthy baseline, providing a falsifiability check on whether the conditioning input influences generator output.

IV. Experimental Setup

Evaluation protocols. We evaluate under two split modes on each benchmark. *Stratified* (baseline): 70/15/15 stratified random split, seed 42. *Chronological* (deployment-realistic proxy): on AI4I, windows are sorted by `window_start` and split 70/15/15 in temporal order with a K=10 gap buffer between partitions to prevent look-ahead overlap; on CMAPSS, the split is engine-level (engines 1–70 train, 71–85 validation, 86–100 test) with an engine-aware windowing function that prevents windows from crossing unit boundaries, and the scaler is fit on training engines only.

Multi-seed robustness. We repeat the stratified AI4I pipeline across five seeds $\{42, 123, 999, 2026, 7\}$, each with a full 60-trial Optuna sweep, and report mean $\pm$ 95% CI.

Forecasting baselines. For the TFT evaluation we compare against two trivial forecasters: *Persistence* (last observed value copied forward) and *Moving Average* (mean of the W=20-cycle input window, repeated across the horizon), both evaluated under the same test windows with a $\pm 1\sigma_{\text{train}}$ uncertainty band.

Metrics. ROC-AUC, Average Precision (AP), F1, Recall, Precision; FTEPH coverage at $h_{\min} \in \{5, 10, 20\}$; TFT RMSE/MAE/80%-band coverage; simulation consistency.

Baselines on AI4I: LSTM [4], PatchTransformer with TS-TCC pretraining and focal-loss fine-tuning, and β-VAE reconstruction-error detection [6]. On CMAPSS we compare against PatchTransformer under the same horizon-based formulation.

V. Results

A. Detection and Early Warning on AI4I 2020 (Stratified)

Table I reports detection performance under the stratified protocol. The horizon-based XGBoost reaches ROC-AUC $= 0.9918$, AP $= 0.9725$, F1 $= 0.9245$, outperforming the PatchTransformer baseline (0.9129, 0.7964, 0.7337) by substantial margins. All 330 failure events are detected with $\geq$5 cycles of lead time; the overall mean horizon is 33.2 cycles (Table II). The per-type ordering is mechanistically interpretable: HDF (47.7 cycles) is longest because thermal imbalance builds gradually via TempDiff; TWF (22.3 cycles) is shortest because wear-induced fracture is inherently abrupt.

TABLE I
AI4I 2020 Binary Detection (K=10, W=20, stratified split).

Metric	XGBoost (ours)	PatchTransformer
ROC-AUC	**0.9918**	0.9129
Avg. Precision	**0.9725**	0.7964
F1-score	**0.9245**	0.7337
Recall	**0.9569**	0.7871
Precision	**0.8942**	0.6871

TABLE II
FTEPH on AI4I 2020 (stratified split; L=5K lookback).

Type	N	Mean h	Coverage at $h_{\min}$		
			≥ 5	≥ 10	≥ 20
HDF	115	47.7	100%	99%	98%
PWF	92	27.1	100%	72%	55%
OSF	80	25.3	100%	70%	49%
TWF	43	22.3	100%	67%	40%
Overall	330	**33.2**	**100%**	80%	67%

B. Multi-Seed Robustness

Table III reports detection and diagnosis confidence intervals across five seeds. The seed-42 numbers reported in Table I lie inside all 95% CIs, confirming they are representative rather than optimistic outliers. The largest seed-to-seed variation is on Recall (std $= 0.025$), driven primarily by which threshold the validation F1-sweep selects (range 0.25–0.55 across seeds). Diagnosis macro-F1 is stable across all classes; the per-type ordering HDF $>$ PWF $>$ OSF $>$ TWF is preserved across all seeds.

TABLE III
Multi-Seed Robustness on AI4I 2020 (5 seeds: {42, 123, 999, 2026, 7}), stratified split.

Metric	Mean	95% CI ($\pm$)
Binary Detection		
ROC-AUC	0.9868	0.0052
AP	0.9525	0.0244
F1	0.9130	0.0177
Precision	0.8985	0.0198
Recall	0.9283	0.0311
Diagnosis (RNF excluded)		
Macro F1	0.9556	0.0113
Weighted F1	0.9600	0.0126
F1 (HDF)	0.9768	0.0203
F1 (PWF)	0.9626	0.0213
F1 (OSF)	0.9545	0.0160
F1 (TWF)	0.9286	0.0194

C. Evaluation Protocol Sensitivity

Table IV reports detection and FTEPH metrics under both split modes on both benchmarks.

TABLE IV
Stratified vs. chronological evaluation. Chronological is the deployment-realistic proxy. AI4I lacks a true time axis; its chronological result is a row-order proxy that strips window-overlap leakage and should be read as a lower bound. CMAPSS uses engine-level chronological splitting, which exposes the model to genuinely unseen engines.

Metric	AI4I 2020		CMAPSS FD001	
	Strat.	Chrono	Strat.	Chrono
ROC-AUC	0.992	0.767	0.999	0.992
Avg. Precision	0.973	0.380	0.985	0.870
F1	0.925	0.518	0.944	0.770
Recall	0.957	0.569	0.960	0.813
Precision	0.894	0.475	0.929	0.731
FTEPH mean (cyc.)	33.2	33.0	46.2	46.5
Coverage $h\geq$5	100%	99%	100%	100%
Coverage $h\geq$10	80%	80%	98%	98%
Coverage $h\geq$20	67%	68%	95%	95%
TFT RMSE (overall)	0.137	0.137	0.126	0.128
TFT 80%-band coverage	86.5%	86.5%	78.3%	78.2%

The pattern is qualitatively different on the two benchmarks. On AI4I, removing window-overlap leakage costs 22 AUC points ($0.992 \rightarrow 0.767$) and 41 F1 points ($0.925 \rightarrow 0.518$): the stratified number was substantially inflated by the same window appearing in train and test up to a single-cycle shift. On CMAPSS, where engines are independent and the chronological protocol exposes the model to genuinely unseen units, the same effect costs only 0.7 AUC points ($0.999 \rightarrow 0.992$) and 16 F1 points ($0.944 \rightarrow 0.770$); the full ROC and PR curves under this protocol are shown in Fig. 1. The residual F1 drop reflects realistic domain shift between training and held-out engines.

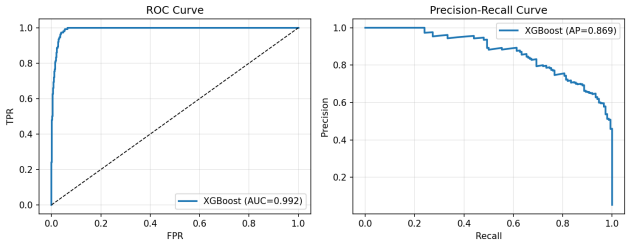


Fig. 1. ROC and Precision-Recall curves under engine-level chronological evaluation on CMAPSS FD001 (AUC = 0.992, AP = 0.870). The PR curve is the more informative metric under the 5.3% positive rate of the chronological test set.

Critically, *FTEPH is essentially invariant* across split modes. On AI4I, mean lead-time changes by 0.2 cycles and event coverage at $h_{\min}$=5 changes by <1 percentage point. On CMAPSS, mean lead-time changes by 0.3 cycles and coverage at every threshold is identical. This dissociation is the key methodological finding: the *operational* early-warning signal that maintenance teams act on is robust to evaluation protocol, whereas the *instantaneous window-classification* accuracy that traditional reports lead with is partly not. Under chronological evaluation, CMAPSS FTEPH of 46.5 cycles with 95% coverage at $h \geq 20$ across 66 test events represents a strong operational signal on genuinely unseen engines.

D. Diagnosis, TFT Forecasting, and Simulation

Diagnosis. Under stratified evaluation the multiclass diagnosis head reaches macro-F1 = 0.9591 (Table V). Under chronological evaluation the AI4I test set contains no HDF samples and the model never predicts TWF, collapsing the macro-F1 to 0.51; the weighted F1 of 0.59 (averaging only classes with non-zero test support: OSF = 0.73, PWF = 0.80) is the comparable figure and is reported with this caveat.

TFT vs. forecasting baselines. Table VI compares the TFT against persistence and moving-average forecasters. The TFT reduces RMSE by 27.3–36.8% on both datasets. Its learned q_{10}–q_{90} interval achieves 78–87% empirical coverage against an 80% nominal target, compared with 53–72% for the baselines' static $\pm 1\sigma_{\text{train}}$ bands. Note that the band constructions differ methodologically: the TFT uses adaptive conditional intervals while baselines use fixed-width training statistics; the coverage comparison therefore understates the TFT's advantage in adaptive uncertainty. Fig. 2 shows representative sensor trajectories with the 80% band on CMAPSS, confirming that the true future consistently falls within the predicted interval.

Decision-support simulation. Table VII reports closed-loop consistency on AI4I. Healthy-seed controls confirm that failure-type conditioning influences generator output: mean healthy-seed risk varies from 0.069 (TWF, neutral response) to 0.236 (HDF, strongest elevation) across conditioning labels, providing evidence that the conditioning input is consumed by the generator. On CMAPSS (single failure class) the healthy-seed control is structurally uninformative—a single class cannot produce per-type variation—and should not be read as a generator failure. The CMAPSS pre-failure seed yields consistency = 0.425 with detector threshold re-crossing at simulated step 16 (Fig. 3). HDF has no entry in Table VII because no test window satisfied the strict true-positive, high-confidence seed criterion; this is reported rather than imputed.

TABLE V
DIAGNOSIS (AI4I, STRATIFIED) AND PER-TYPE TFT PERFORMANCE (AI4I, CHRONOLOGICAL SPLIT).

Diagnosis	Value	TFT type	RMSE	MAE	80% Cov.
Macro F1	0.9591	HDF	0.110	0.067	86%
Weighted F1	0.9664	PWF	0.133	0.076	86%
F1 (HDF)	0.9908	OSF	0.147	0.085	87%
F1 (PWF)	0.9683	TWF	0.170	0.095	87%
F1 (OSF)	0.9579	**Overall**	**0.137**	**0.079**	**86.5%**
F1 (TWF)	0.9195				

TABLE VI
TFT VS. FORECASTING BASELINES (CHRONOLOGICAL SPLIT). BAND TYPES DIFFER; SEE TEXT.

Dataset	Forecaster	RMSE	MAE	RMSE Δ vs TFT
AI4I	Persistence	0.189	0.114	+27.3%
	Moving Average	0.173	0.108	+20.6%
	TFT (ours)	**0.137**	**0.079**	—
CMAPSS	Persistence	0.203	0.140	+36.8%
	Moving Average	0.176	0.133	+27.1%
	TFT (ours)	**0.128**	**0.089**	—

TABLE VII
SIMULATION TRAJECTORY CONSISTENCY ON AI4I 2020 (HEALTHY BASELINE RISK = 0.100; 40-STEP SIMULATION; CHRONOLOGICAL SPLIT). HDF: NO QUALIFYING SEED WINDOW IN TEST SET.

Type	Mean Risk	Peak Risk	Risk@K	Consistency
PWF	0.305	0.814	0.341	0.232
OSF	0.477	0.987	0.957	0.422
TWF	0.380	0.918	0.853	0.315
HDF		— (no qualifying seed) —		
CMAPSS	0.437	0.994	0.120	0.425

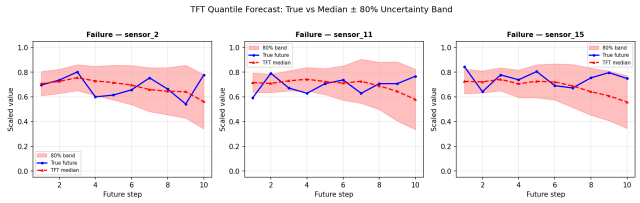


Fig. 2. TFT quantile forecast on CMAPSS FD001 (chronological split): true future trajectory (blue) vs. q_{50} median (dashed red) with 80% uncertainty band (q_{10}–q_{90}, shaded) across three representative sensors. The true trajectory remains within the band in the majority of steps, consistent with the 78.2% empirical coverage reported in Table IV. The widening band at longer horizons reflects growing uncertainty, a property fixed-width baseline bands cannot express.

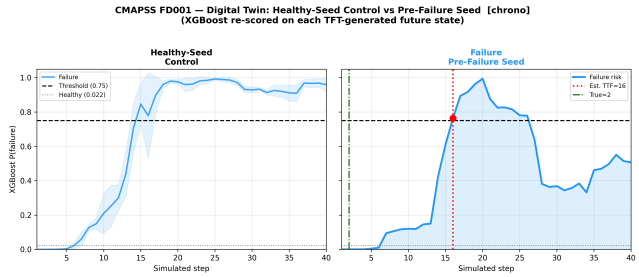


Fig. 3. CMAPSS decision-support simulation (chronological split). *Left*: mean detector risk ± std across 5 healthy seeds—trajectories remain below the decision threshold, confirming the generator does not produce failure-like outputs for healthy-conditioned inputs. *Right*: pre-failure seed—simulated risk crosses the threshold at step 16 (estimated TTF), with true failure at offset 2 from the seed.

E. Component Ablation

Table VIII reports a four-row ablation on AI4I (stratified). Three findings stand out. **(i)** Temporal sliding-window statistics are dominant: removing them costs $\Delta\text{AUC} = -0.117$, $\Delta\text{AP} = -0.199$, $\Delta\text{F1} = -0.159$, confirming that early failure signatures are encoded in window-level drifts and slopes rather than instantaneous sensor values. **(ii)** Physics features add a smaller but consistent margin ($\Delta\text{AUC} = -0.009$, $\Delta\text{AP} = -0.020$). **(iii)** The TFT does not affect detection metrics by design: the detector and forecaster operate in parallel, and the flat row for "No TFT" documents that the TFT contributes generative coherence (Table VII) rather than classification accuracy.

TABLE VIII
COMPONENT ABLATION ON AI4I 2020 (STRATIFIED). Δ VALUES ARE ABSOLUTE VS. FULL SYSTEM.

Variant	AUC	AP	F1	ΔAUC	ΔAP	ΔF1
Full system	**0.9918**	**0.9725**	**0.9245**	—	—	—
No physics	0.9828	0.9525	0.9205	−0.009	−0.020	−0.004
No temp. stats	0.8748	0.7735	0.7655	−0.117	−0.199	−0.159
No TFT	0.9918	0.9725	0.9245	0.000	0.000	0.000

F. Cross-Dataset Generalization: CMAPSS FD001

To verify that the methodology—not dataset-specific tuning—drives performance, we apply the identical pipeline to CMAPSS FD001 under both evaluation protocols. The only change is dropping AI4I-specific physics terms (not derivable from anonymized CMAPSS sensors) and skipping diagnosis (single failure mode).

Under chronological evaluation (the deployment-realistic proxy), the system reaches ROC-AUC = 0.992, F1 = 0.770, and a CMAPSS FTEPH of 46.5 cycles mean lead-time across 66 test failure events, with 100% coverage at $h_{\min}$=5 and 95% at $h_{\min}$=20. These results are operationally meaningful because CMAPSS has a genuine time axis and the engine-level chronological split constitutes a true held-out scenario. The PatchTransformer gap is much smaller on CMAPSS (0.7 AUC points) than on AI4I (7.9 points), consistent with CMAPSS providing longer continuous degradation trajectories from which a deep model can extract useful structure.

VI. DISCUSSION

Framework applicability across failure regimes. AI4I and CMAPSS represent opposing extremes: point-like, abrupt failures versus smooth monotonic degradation. The horizon-based label is compatible with both because it is defined purely from failure timestamps; the temporal sliding-window statistics then surface whatever precursor pattern the regime provides. The ablation result (Table VIII) confirms this: temporal statistics dominate on AI4I despite the abrupt regime, because the *window-level* change in slope or variance is detectable even when the raw sensor values are not individually anomalous. On CMAPSS, where degradation is smooth, the same statistics capture drift slopes naturally. The gap between XGBoost and deep baselines narrows on CMAPSS (0.7 vs. 7.9 AUC points), consistent with deep models gaining traction only when long-range temporal structure is present.

Why β-VAE fails on AI4I. Failure events are point-like and immediately preceded by sensor values within the nominal envelope; reconstruction error cannot exceed the detection threshold until failure has occurred. The horizon-based reformulation bypasses this by using temporal proximity to failure as the label.

The FTEPH–AUC dissociation. The stratified-vs-chronological experiment reveals that FTEPH is split-invariant while AUC is not. This is methodologically important: the *operational utility* of the system (how early it warns before each event) is robust to the evaluation protocol, while the raw binary classification accuracy partly reflects window-overlap leakage. Papers reporting only AUC under stratified splits may substantially overestimate performance on temporally correlated streams.

Tabular vs. deep models. The horizon-aware XGBoost outperforms deep baselines on AI4I because failures are abrupt with minimal temporal structure in the 20 preceding cycles; the gap narrows on CMAPSS, where longer smooth degradation trajectories favor sequence models. Fig. 4 provides a visual explanation: UMAP projection of the PatchTransformer's CLS token embeddings shows zero class separation on AI4I, confirming that the deep model's temporal inductive bias is structurally mismatched to this failure regime.

Simulation coherence. The consistency score provides a falsifiability check: a forecaster ignoring the failure-type condition would produce risk near the healthy baseline (≈0.10 on AI4I). Observed consistency scores (0.23–0.42 on AI4I; 0.425 on CMAPSS) provide empirical evidence that TFT-generated trajectories are recognized as failure-like by an independently trained detector that never shared parameters with the generator. All consistency scores fall below 0.5, consistent with conservative q_{50} forecasting: the median smooths the predictive tail, placing trajectories in a credible degradation zone rather than at saturated certainty.

Limitations. The TFT q_{50} smooths sharp transitions and underperforms on TWF vs. HDF (RMSE 0.170 vs. 0.110).

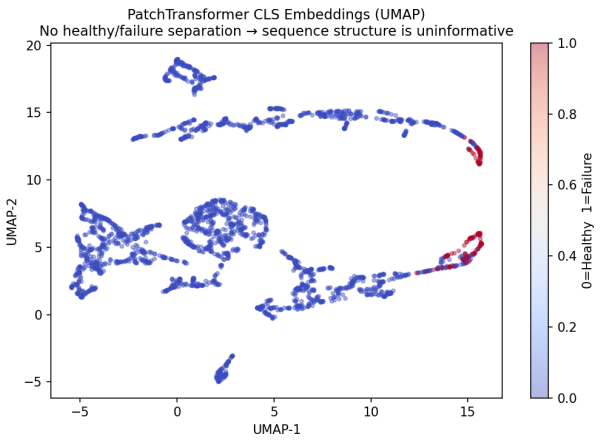


Fig. 4. UMAP projection of PatchTransformer CLS token embeddings on AI4I 2020. Failure samples (red) are indistinguishable from healthy samples (blue) in the learned representation space, confirming that temporal sequence structure is uninformative for AI4I's abrupt, point-like failure regime. This structural mismatch explains why the horizon-aware XGBoost detector outperforms the deep baseline by 7.9 AUC points.

AI4I lacks a true time axis; its "chronological" result is a row-order proxy and should not be equated with engine-level chronological evaluation on CMAPSS. Diagnosis weighted F1 under chronological AI4I (0.59) is not directly comparable to stratified (0.96) because the test set contains zero HDF samples and the model never predicts TWF; per-class support should be consulted. Healthy-seed controls on CMAPSS are structurally limited by single-class conditioning; the AI4I multi-type control is the primary falsification test. Finally, this work is validated on publicly available benchmarks; deployment to real industrial environments would require additional calibration and operational validation.

VII. Conclusion

We presented *IoT-DriftGen*, a horizon-based predictive-maintenance framework whose central finding is a methodological one: FTEPH—the operationally meaningful early-warning metric—is essentially invariant to evaluation protocol (change <1 cycle on both benchmarks) even when raw classification AUC drops by 22 points on AI4I. This FTEPH–AUC dissociation argues that event-level lead-time metrics should be reported alongside classification scores in PdM evaluations, and that stratified AUC on temporally correlated streams may substantially overstate deployment performance.

The detection backbone combining physics-informed features with sliding-window temporal statistics achieves ROC-AUC $= 0.9868 \pm 0.0052$ across five seeds, with ablation confirming that temporal statistics are the dominant contributor. A compact TFT adaptation augments the pipeline with calibrated uncertainty bands (20–37% RMSE reduction over trivial baselines) and failure-type-conditioned trajectory simulation, enabling uncertainty-aware decision support. Together, these components advance predictive maintenance evaluation from single-timestep alarms toward an interpretable, horizon-aware framework validated under realistic temporal conditions.

Acknowledgment

The authorship team would like to acknowledge the vision, support and guidance of the IEEE Industrial Electronics Society in conducting the Generative AI Hackathon under the leadership of Daswin De Silva and Lakshitha Gunasekara.

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